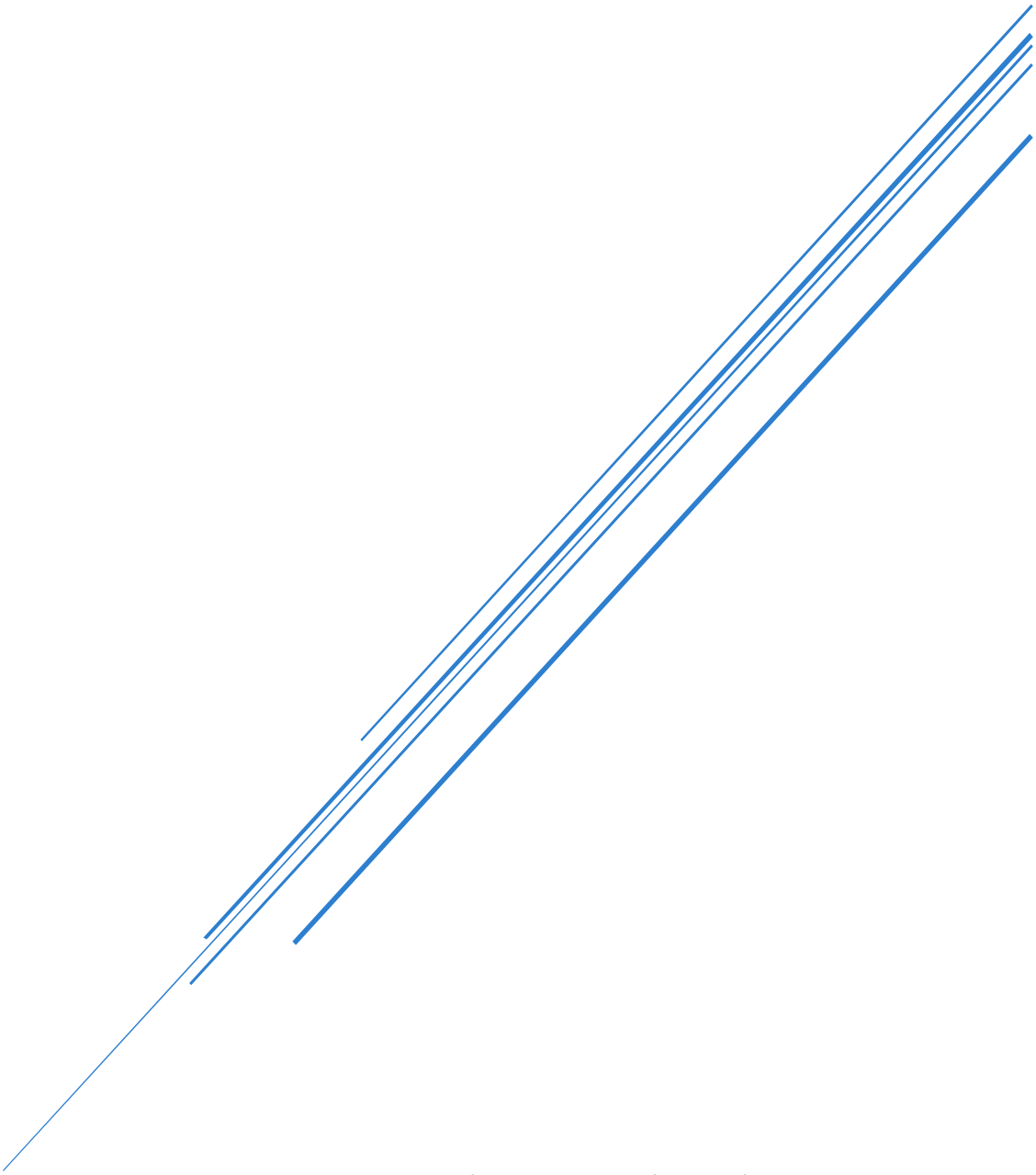
LAB 9 ANSWERS REPORT

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Table of Contents

Exercises ***2***

Question 1 ..... **2**

Question 2 ..... **2**

a) Is it possible to travel to Darwin on the *Quokka Express* Network? 2
b) Is it possible to travel to Hobart on the *Quokka Express* Network? 2
c) Find a path to Melbourne 2

Question 3 ..... **2**

Question 4 ..... **2**

a) Calculate the total travel time on the Quokka Express transport network from Adelaide to Brisbane 2
b) Is it quicker to travel to Darwin or Brisbane via the Quokka Express? 3
c) What is the time difference between travelling from Adelaide to Sydney compared to travelling Adelaide to Melbourne on the Quokka Express? 3

Question 5 ..... **3**

Question 1 ..... ***3***

Generate a sentence using the grammar.pl knowledge base using Prolog **3**

Question 2 ..... ***4***

Parse the sentence "the big cat chases a mouse" using the grammar.pl knowledge base and Prolog **4**

Question 3 ..... ***4***

List all possible noun phrases that can be generated using the context-free grammar specified in the grammar.pl knowledge base **4**

Question 4 ..... ***4***

Check to see if "a small dog sees the cat" is a valid sentence according the context-free grammar described in grammar.pl **4**

Question 5 ..... ***4***

Extend the context-free grammar described in grammar.pl to include prepositional phrases. Once the grammar is extended use it generate a sentence with a prepositional phrase **4**

Exercises

Using the Quokka.pl prolog file as a knowledge base here are the queries and answers.

Question 1

Done in the Quokka.pl file.

Question 2

a) Is it possible to travel to Darwin on the *Quokka Express* Network?

Query:
travel(_, darwin, _).

Result:
true

b) Is it possible to travel to Hobart on the *Quokka Express* Network?

Query:
travel(_, hobart, _).

Result:
false.

c) Find a path to Melbourne

Query:
path(adelaide, melbourne, Path, Time).

Result:
Path = [adelaide, canberra, melbourne],
Time = 31.

Question 3

The above results make sense since there is a direct path for travel between Adelaide and Darwin whereas Hobart is not even in the network. Using the placeholder Path allows us to get both the path and total travel time from Adelaide to Melbourne.

Question 4

a) Calculate the total travel time on the Quokka Express transport network from Adelaide to Brisbane

Query:
path(adelaide, brisbane, _, Time).

Result:
Time = 52

b) Is it quicker to travel to Darwin or Brisbane via the Quokka Express?

Query:
path(adelaide, brisbane, _, BrisbaneTime), path(adelaide, darwin, _, DarwinTime), (DarwinTime < BrisbaneTime -> write('Darwin is Quicker'); write('Brisbane is Quicker')).

Result:
Darwin is Quicker
BrisbaneTime = 52,
DarwinTime = 44 .

c) What is the time difference between travelling from Adelaide to Sydney compared to travelling Adelaide to Melbourne on the Quokka Express?

Query:
path(adelaide, sydney, _, SydneyTime), path(adelaide, melbourne, _, MelbourneTime), TimeDifference is abs(SydneyTime - MelbourneTime).

Result:
SydneyTime = 27,
MelbourneTime = 31,
TimeDifference = 4 .

Question 5

Query:
findall(Time, travel(_, _, Time), Times), max_list(Times, MaxTime).

Result:
Times = [44, 22, 28, 30, 5, 9, 44, 22, 28|...],
MaxTime = 44.

Question 1

Generate a sentence using the grammar.pl knowledge base using Prolog

Query:
phrase(sentence, S).

Result:
S = [the, cat, chases, the, cat] .

When I keep hitting ‘;’ it keeps doing different sentences but ends up looping after expending all options.

Question 2

Parse the sentence “the big cat chases a mouse” using the grammar.pl knowledge base and Prolog

Query:
phrase(sentence, [the, big, cat, chases, a, mouse]).

Result:
true .

Question 3

List all possible noun phrases that can be generated using the context-free grammar specified in the grammar.pl knowledge base

Query:
findall(N, phrase(noun_phrase, N), Noun_Phrases).

Result:
Noun_Phrases = [[the, cat], [the, dog], [the, mouse], [a, cat], [a, dog], [a, mouse], [the, big|...], [the|...], [...|...]|...].

Question 4

Check to see if “a small dog sees the cat” is a valid sentence according the context-free grammar described in grammar.pl

Query:
phrase(sentence, [a, small, dog, sees, the, cat]).

Result:
true .

Question 5

Extend the context-free grammar described in grammar.pl to include prepositional phrases. Once the grammar is extended use it generate a sentence with a prepositional phrase

Grammar was updated accordingly in the grammar.pl file.

Query:
phrase(prepositional_phrase, P).

Result:
P = [on, the, cat] .